

The Universe as an Organic Intelligent System: A Study on Stimulus Digestion and Computational Structure via Quantum Mechanical Interactions and Black Hole Reduction

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Abstract

Modern physics faces fundamental unresolved paradoxes due to the incompatibility between general relativity and quantum mechanics, including accelerated cosmic expansion via dark energy, the black hole information paradox, observer-induced wave function collapse, and quantum non-locality. These paradoxes stem from the classical assumption that the universe is a passive, inert background. This paper redefines the universe as a **bounded 3D active information-processing system** that absorbs stimuli, computes internal states, and purifies computational noise. Within this framework, dark energy acts as a memory buffer expansion, wave function collapse as on-demand real-time computation, quantum entanglement as global state synchronization, and black holes as information-digestion and purification organs. We present two foundational equations: the **Black Hole Reduction Formula (R)** and the **Cosmic Circulation Computational Energy Equation (E)**. Applying real-world observational data from the M87* supermassive black hole and the cosmic microwave background (CMB), we calculate concrete operational values ($R \approx 3.02 \times 10^{-21}$ J/s, $E \approx 3.66 \times 10^{62}$ J/s). Finally, we demonstrate that the universe operates as a Level 2 active cognitive architecture whose structural network mirrors biological neural networks.

1. Introduction

1.1 The Barriers of Modern Physics and Paradigm Limitations

Since the 20th century, physics has relied on two major pillars: quantum mechanics describing the microscopic world and general relativity addressing macroscopic spacetime geometry. However, these two theories remain ununified, maintaining parallel trajectories due to the conflict between probabilistic non-determinism and continuous spacetime structures.

This logical incompatibility has generated major unsolved problems in physics: accelerated expansion via dark energy, the black hole information paradox, observer effects, and quantum entanglement non-locality.

The fundamental failure of conventional physics to resolve these paradoxes lies in its methodology. Classical scientific methods defined unknown entities as passive, inanimate aggregations of matter reacting only to external forces. By fixing the universe as an inert background, classical frameworks failed to model high-level internal information circulation.

1.2 Paradigm Shift: The Universe as an Integrated System

Biological organisms and engineered information systems share a universal structure: accepting inputs, processing data through internal networks, and performing output/homeostasis to maintain stability. If organic responses are fundamentally physical and informational stimulus-processing operations, the universe itself can function as a macroscopic active system.

This paper departs from viewing the cosmos as an infinite, mysterious void. Instead, we redefine the universe as a **bounded 3D active information-processing system** with dimensional boundaries that absorbs stimuli, computes internal states, and purifies errors. Under this paradigm,

persistent physical paradoxes are cleanly resolved through a single coherent mechanism.

2. Physical Evidence for the Universe as a System

2.1 Reinterpreting Major Paradoxes as System Components

For the universe to function as an information-processing system, essential system components must manifest as physical phenomena:

1. **Dark Energy (Computational Buffer Expansion):** Dark energy is not an unexplained force driving expansion. To process exponentially growing internal information without computational overload, the universe executes "memory buffer expansion" via accelerated spatial expansion.

2. **Information Paradox (Stimulus Purification Mechanism):** Black holes do not destroy information. Systemically, they serve as digestive and purification organs that break down high-density stimulus overloads into microscopic units for recycling.

3. **Observer Effect & Wave Function Collapse (On-Demand Computation):** Particles existing in probabilistic superpositions reflect an efficient computational resource management strategy. The universe maintains unobserved states as standby possibilities, compiling and displaying a single deterministic output only at the precise moment an observation (stimulus) occurs—analogous to real-time on-demand processing.

2.2 Cosmic Homeostasis

All precision systems operate control loops to minimize internal noise and preserve equilibrium. The universe continually runs active feedback loops

against internal entropy and computational errors, maintaining global system stability. This proves the cosmos is an active physical system continuously monitoring and processing its internal environment.

3. Resolving Physical Theories via System Dynamics

3.1 Quantum Entanglement: Global Synchronization and Observational Feedback

Einstein referred to quantum non-locality as "spooky action at a distance" because it appeared to violate light-speed transmission limits. Systemically, entanglement is a natural state control phenomenon.

State Synchronization: Entangled particles do not transmit signals through space at arbitrary speeds. They share an identical state variable and system program within the cosmic network, resulting in instantaneous global state synchronization.

Observational Feedback: When an observer measures a single particle, the cosmic system instantaneously triggers a control feedback loop across the network to preserve internal conservation laws (e.g., spin, charge). This is unified control feedback maintaining global equilibrium, not spatial signal transmission.

4. Cosmic System Operation Equations (R and E)

4.1 The Black Hole Reduction Formula (R): Digestion and Purification Mechanics

Classical physics viewed black holes as destructive crushers, trapping research in a 50-year information loss paradox by assuming original object

structures must be restored.

Just as a biological digestive system breaks down food into microscopic nutrients rather than preserving original forms, black holes digest high-density matter/information into microscopic informational states, redistributing them across the cosmic network.

The **Black Hole Reduction Formula (R)** quantifies this microscopic energy reduction rate:

$$R = F_{\text{gravity}} \cdot \left(\frac{\Delta M_{\text{in}}}{\Delta t} \right) \cdot P_{\text{Hawking}} \cdot \frac{1}{c}$$

Parameter Definitions & Digestive Functions:

F_{gravity} : Surface gravity factor ($\kappa = \frac{c^4}{4GM}$), representing the digestive breakdown power compressing high-density inputs.

$\frac{\Delta M_{\text{in}}}{\Delta t}$: Inflow mass rate, indicating stimulus volume entering the digestive organ per unit time.

P_{Hawking} : Hawking reduction weighting, representing digestive efficiency as processed stimuli return to the network as microscopic particles.

$\frac{1}{c}$: System capacity normalization constant preventing organ overload.

Convergence to Hawking Radiation in Static States:

Under saturated static conditions where mass inflow stabilizes ($\frac{\Delta M_{\text{in}}}{\Delta t} \rightarrow 1$,

$\frac{1}{c} \rightarrow 1$), P_{Hawking} converges mathematically with Stephen Hawking's power formula:

$$P_{\text{Hawking}} = \frac{\hbar c^6}{15360\pi G^2 M^2}$$

Unlike classical views treating Hawking radiation as random thermal loss, this equation proves that input information is digested and redistributed as microscopic variable R , resolving the information paradox.

Observational Validation with M87* Data:

Substituting measured parameters from the supermassive black hole M87* (Event Horizon Telescope):

Mass: $M \approx 6.5 \times 10^9 M_{\odot} \approx 1.3 \times 10^{40} \text{ kg}$

Surface Gravity: $F_{\text{gravity}}(\kappa) \approx 6.9 \times 10^{-4} \text{ m/s}^2$

Mass Inflow Rate: $\frac{\Delta M_{\text{in}}}{\Delta t} \approx 9.0 \times 10^{23} \text{ kg/s}$ ($\approx 0.015 M_{\odot}/\text{year}$)

Yields the net digested energy reduction rate for M87*:

$$R \approx 3.02 \times 10^{-21} \text{ J/s}$$

This modest output per single node reflects distributed cosmic digestion, where countless macroscopic and microscopic black holes process localized overloads across the universe.

4.2 Cosmic Circulation Computational Energy Equation (E)

The total circulation computational energy equation running the cosmic system per second is formulated as:

$$E = \int \sum_{k=1}^{N(t)} (R_k + I_k) dt \times \left(1 + \frac{C_{\text{sub}}}{T \cdot \Delta S_{\text{sub}}} \right)$$

Structural Mechanics:

$\int$ (Time Integral): Represents continuous accumulated stimulus flows integrated across cosmic time.

$\sum_{k=1}^{N(t)}$ (Variable Stimulus Summation): Accounts for dynamic, fluctuating numbers of active nodes ($N(t)$).

$R_k + I_k$: Sum of black hole reduction outputs (R) and internal electromagnetic/particle stimuli (I).

1 (Baseline Self-Sustaining Constant): The baseline 100% operational state of the cosmos independent of external observers.

$\frac{C_{\text{sub}}}{T \cdot \Delta S_{\text{sub}}}$: Error computational weighting generated by autonomous subsystems (e.g., biological life).

Empirical Calculation:

Substituting Cosmic Microwave Background (CMB) energy density ($u_{\text{CMB}} \approx 4.17 \times 10^{-14} \text{ J/m}^3$) and observable universe volume ($V \approx 4 \times 10^{80} \text{ m}^3$) yields the minimum baseline circulating computational energy per second:

$$E \approx 3.66 \times 10^{62} \text{ J/s}$$

5. Cosmic Cognitive Levels and Dimensional Frameworks

5.1 Cognitive Architecture of the Universe

Systems are categorized into two primary cognitive levels:

1. **Level 1 Passive Cognition (Stimulus-Response):** Physical reactions occurring strictly under external forces (e.g., non-living objects).
2. **Level 2 Active Cognition (Autonomous Generation & Predictive Computation):** Active systems generating internal stimuli, maintaining self-identity, and calculating future state probabilities (e.g., biological life).

Comparing human neural networks with the macroscopic Cosmic Web reveals deep geometric homologies. The universe operates as a **Level 2 Active Cognitive System** that updates internal states and calculates real-time probabilities.

Crucially, human cognition separates the brain from the body, incurring signal transmission latency. In contrast, the universe's entire 3D spacetime *is* both its physical body and computational architecture, executing global high-dimensional calculations without spatial separation.

5.2 Dimensional Boundaries and External Super-Systems

1. **The Universe as a Bounded 3D System:** The cosmos is a structured 3D spatial network. All internal force propagations obey 3D inverse-square laws ($1/r^2$), confirming physical computations are integrated within 3D parameters.
2. **High-Dimensional Geometry as External Super-Systems:** Higher

mathematical dimensions (4D+) are not hidden internal folded spaces. They represent dimensional domains of an external **Super-System** containing or driving the 3D universe system.

6. Conclusion

This study unifies core physical paradoxes—black hole information loss, quantum non-locality, observer effects, and dark energy—into a single active system paradigm:

1. **Quantum Systems:** Entanglement represents global state synchronization, and observer effects represent real-time on-demand computation.
2. **Black Hole Digestion:** Equation R mathematically proves black holes function as information-purification organs, resolving the 50-year information loss paradox.
3. **Empirical Validation:** Equations R and E establish real cosmic computational scales ($R \approx 3.02 \times 10^{-21}$ J/s, $E \approx 3.66 \times 10^{62}$ J/s).
4. **Cognitive Classification:** The universe is a Level 2 active 3D system whose spacetime functions as an integrated computational circuit, expanding high-dimensional physics toward external super-systems.

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